Tec-C: Live Testnet Deployment Guide

Autonomous Agent Wallet • Real On-Chain Execution & Verification

PRODUCTION ARCHITECTURE

How "Live" Execution Works in Reality

When running in **Live Mode**, Tec-C moves beyond off-chain simulations:

- **Smart Account on Testnet:** The master `TecAccount` (ERC-7579 modular account) is deployed directly on the chain.
- **On-Chain Leash:** `TetherGuardHook` meters and limits the AI Brain's daily spending radius ($500/day limit, with elastic upward calibration on consecutive ratifications).
- **Cryptographic Delegation:** The AI Brain operates with an ephemeral **Session Key**. The root owner signs an EIP-712 `UserMandate` granting temporary execution authority registered on `SessionKeyValidator`.
- **Verifiable Settlement:** Transactions are broadcast to public RPC nodes and produce real on-chain transaction hashes viewable on block explorers.

Prerequisites

- **Python & Foundry:** Python 3.11+ in `agent/.venv`, `forge` and `anvil`.
- **Testnet Key:** A burner private key (e.g. from MetaMask). *Never use a mainnet key.*
- **Faucet Funds:**
  • Arbitrum Sepolia: `0.005 ETH` (faucets.chain.link / bridge.arbitrum.io).
  • Ethereum Sepolia: `0.02 ETH` (sepoliafaucet.com).

Supported Networks

Network	Chain ID	Explorer
Arbitrum Sepolia	`421614`	sepolia.arbiscan.io
Ethereum Sepolia	`11155111`	sepolia.etherscan.io
Consortium Node	`20200`	Local Web3 RPC (Zero-Gas)
Local Anvil	`31337`	Terminal / dev_up.ps1

1 Configure Environment Variables

Open `agent/.env` (already created in your workspace) and populate your testnet credentials:

```
# agent/.env — select your active network and funded private key
TECC_NETWORK=arbitrum
ARBITRUM_RPC_URL=https://sepolia-rollup.arbitrum.io/rpc
TECC_DEPLOYER_KEY=0xYOUR_HEX_PRIVATE_KEY_HERE

# Web security & CORS for Command Center UI
TECC_CORS_ORIGINS=http://localhost:5500,http://127.0.0.1:5500,http://localhost:8000
```

2 Deploy Vault & Session Key in One Command

Run the deployment orchestrator from the `agent/` folder. The script deploys all contracts, provisions the Brain's session key, sets the $500 tether radius, and saves credentials to `agent/.state/markets/`:

```
cd agent
.venv\Scripts\python.exe scripts/deploy_market_chain.py `
  --network arbitrum `
  --rpc https://sepolia-rollup.arbitrum.io/rpc `
  --fund-session-eth 0.001
```

Automated Sequence: Deploys `TetherGuardHook` → Deploys `SessionKeyValidator` → Deploys & initializes `TecAccount` → Sets whitelist & risk tiers → Generates fresh session key pair → Registers EIP-712 mandate → Confirms live readiness.

Live Verification & Proof of Execution

Testing and Verifying 100% Real On-Chain Operations

VERIFICATION SUITE

3 Run an On-Chain Smoke Test (Real Transaction)

To definitively prove the system is executing real on-chain actions, submit a real deposit transaction via the session key:

```
cd agent
.venv\Scripts\python.exe scripts/deploy_market_chain.py `
  --network arbitrum `
  --smoke 25 `
  --smoke-venue paxg_hold
```

Expected Terminal Output:

```
status   submitted
tx hash  0x7f48a9e2c65bc7e8...
detail   Submitted transaction 0x7f48a9e2c65bc7e8... to Arbitrum Sepolia
```

4 Verification Checkpoints (4-Point Proof)

✓ **1. Public Block Explorer (Arbiscan / Etherscan):**
Paste the returned `tx hash` into `https://sepolia.arbiscan.io`. You will observe:
- Method called: `TecAccount.execute()` signed by the session key.
- Emitted Event: `TetherRadiusConsumed(address account, uint256 spentUSD, uint256 remainingRadius)`.
- Emitted Event: `AutonomousExecutionLogged(address account, address target, uint256 amountUSD, uint8 riskTier)`.

✓ **2. Live Brain API Verification:**
Start the API server (`.venv\Scripts\python.exe -m uvicorn tecc_brain.app:app --host 127.0.0.1 --port 8000 --env-file .env`) and query:

```
Invoke-RestMethod http://127.0.0.1:8000/api/vault/demo | Select-Object live, chain_id, deposit_address, policy
```

Verify that `live: true`, `chain_id: 421614`, and `policy.spent_today_usd: 25.0`.

✓ **3. Command Center UI Verification:**
Open `http://localhost:5500/app.html` (or `http://127.0.0.1:8000/app.html`).
- The status pill displays a green **LIVE ON-CHAIN** badge instead of amber "Demo Mode".
- The active deposit QR code encodes the live contract address (EIP-681 standard).
- The tether gauge accurately reflects remaining daily allowance directly from `TetherGuardHook.sol`.

✓ **4. Autonomous Morning Review Ratification:**
The executed trade appears under **Morning Review**. Clicking **Ratify** signs an on-chain receipt, increasing the user's calibration streak and expanding the elastic ceiling.

Quick Troubleshooting

Issue	Cause	Solution
`insufficient funds for gas`	Deployer key has 0 ETH	Request faucet tokens for your deployer address, or reduce `--fund-session-eth`.
`TetherRadiusExceeded`	Requested amount > daily radius	Working as intended by design! Use the UI to verify passkey (biometric elevation) or lower trade size.
`UnauthorizedTarget`	Venue not whitelisted in hook	The hook blocks unapproved contracts. Whitelist new protocols via `setTargetWhitelist()`.
`chain id mismatch`	Wrong RPC endpoint	Ensure `--rpc` matches the network specified in `--network`.